Problem 1. From the fact that $k^\alpha = (\frac{\omega}{c}, \vec{k})$ obeys the Lorentz transformation

$$k'^0 = \gamma(k^0 - \beta k^1)$$

Using $k'^0 = \frac{\omega_0}{c}$

$$k^0 = \frac{\omega}{c}$$

$$k^1 = \frac{\omega}{c}\cos\theta$$

we get $\frac{\omega}{\omega_0} = \frac{1}{\gamma(1-\beta\cos\theta)}$

Problem 2.

Two successive Lorentz transformations $\hat{L}_1$ and $\hat{L}_2$ (each given by Eq. 1.130) are equivalent to $\hat{L}_{12}$ given by the matrix product

$$\hat{L}_{12} = \begin{pmatrix} \gamma_1 & -\beta_1\gamma_1 & 0 & 0 \\ -\beta_1\gamma_1 & \gamma_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_2 & -\beta_2\gamma_2 & 0 & 0 \\ -\beta_2\gamma_2 & \gamma_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} \gamma_1\gamma_2(1+\beta_1\beta_2) & -\gamma_1\gamma_2(\beta_1+\beta_2) & 0 & 0 \\ -\gamma_1\gamma_2(\beta_1+\beta_2) & \gamma_1\gamma_2(1+\beta_1\beta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This expression has the same form as prescribed by Lorentz transformation if we impose

$$\gamma_{12}\beta_{12} = (\beta_1+\beta_2)\gamma_1\gamma_2 \quad \text{and} \quad \gamma_{12} = \gamma_1\gamma_2(1+\beta_1\beta_2)$$

Excluding γ_{12} from these equations we find

$$L_{12} = \begin{pmatrix} \gamma_{12} & -\gamma_{12}\beta_{12} & 0 & 0 \\ -\gamma_{12}\beta_{12} & \gamma_{12} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\left.\begin{aligned} \gamma_{12}\beta_{12} &= (\beta_1+\beta_2)\gamma_1\gamma_2 \\ \gamma_{12} &= \gamma_1\gamma_2(1+\beta_1\beta_2) \end{aligned}\right\} \quad \beta_{12} = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2}$$

$$\Rightarrow V_{12} = \frac{V_1+V_2}{1+\frac{V_1V_2}{c^2}}$$

(P4) $A^\alpha = (2,2,0,0)$; $B^\alpha = (2,1,0,0)$

a) In K: $A^\alpha B_\alpha = 2\cdot 2 + 2(-1) = 2.$

In K':
$$\begin{pmatrix} A'^0 \\ A'^1 \\ A'^2 \\ A'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} A^0 \\ A^1 \\ A^2 \\ A^3 \end{pmatrix} = \begin{pmatrix} \gamma A^0 - \gamma\beta A^1 \\ -\gamma\beta A^0 + \gamma A^1 \\ A^2 \\ A^3 \end{pmatrix} = \begin{pmatrix} 2\gamma(1-\beta) \\ 2\gamma(1-\beta) \\ 0 \\ 0 \end{pmatrix}$$

and
$$\begin{pmatrix} B'^0 \\ B'^1 \\ B'^2 \\ B'^3 \end{pmatrix} = \begin{pmatrix} \gamma(2-\beta) \\ \gamma(1-2\beta) \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} B'_0 \\ B'_1 \\ B'_2 \\ B'_3 \end{pmatrix} = \begin{pmatrix} \gamma(2-\beta) \\ -\gamma(1-2\beta) \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow A'^\alpha B'_\alpha = 2\gamma(1-\beta)\cdot\gamma(2-\beta) - 2\gamma(1-\beta)\cdot\gamma(1-2\beta) =$$
$$= 2\gamma^2(1-\beta)\left(2-\beta-(1-2\beta)\right) = 2\gamma^2(1-\beta)(1+\beta) = 2\gamma^2(1-\beta^2) = 2.$$

b) $A^0B^0 + A^1B^1 = 2\cdot 2 + 2\cdot 1 = 6. \quad \Rightarrow \sum_{\alpha=0}^{3} A^\alpha B^\alpha \neq \sum_{\alpha=0}^{3} A'^\alpha B'^\alpha$!

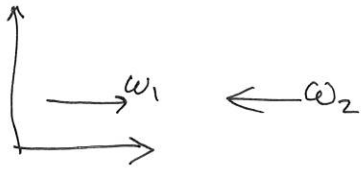
$A'^0B'^0 + A'^1B'^1 = 2\gamma(1-\beta)\cdot[\gamma(2-\beta)+\gamma(1-2\beta)] = 2\gamma^2(1-\beta)\cdot 3(1-\beta) = 6\frac{1-\beta}{1+\beta}$

Brau 1.12

(P3)

$$K^\alpha = \left(\frac{\omega}{c}, \vec{k}\right) = \left(\frac{\omega}{c}, k\right)$$

k_x only



$$L^\alpha{}_\beta = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix}$$

$$\begin{pmatrix} \omega_1'/c \\ k_1' \end{pmatrix} = k_1'^\alpha = L^\alpha{}_\beta k_1^\beta = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \omega_1/c \\ k_1 \end{pmatrix}$$

$$\begin{pmatrix} \omega_2'/c \\ k_2' \end{pmatrix} = k_2'^\alpha = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \omega_2/c \\ -k_2 \end{pmatrix}$$

$$\omega_1' = \gamma(\omega_1 - \beta c k_1)$$
$$\omega_2' = \gamma(\omega_2 + \beta c k_2)$$

let $\dfrac{\omega_1'}{\omega_2'} = 1 = \dfrac{\omega_1 - \beta c k_1}{\omega_2 + \beta c k_2}$

$$\omega_1 - \beta c k_1 = \omega_2 + \beta c k_2$$

$$\omega_1 - \omega_2 = \beta c (k_1 + k_2)$$

$$\Rightarrow \beta c = \frac{\omega_1 - \omega_2}{k_1 + k_2} = \frac{c(k_1 - k_2)}{k_1 + k_2}$$

$$\mathcal{E}' = \hbar\omega_1' + \hbar\omega_2' = \hbar\gamma\left(\omega_1 + \omega_2 - \beta c(k_1 - k_2)\right) =$$

$$= \hbar\gamma\left(\omega_1 + \omega_2 - \frac{(\omega_1 - \omega_2)(k_1 - k_2)}{k_1 + k_2}\right) =$$

$$= \hbar\gamma \frac{(\omega_1 + \omega_2)(k_1 + k_2) - (\omega_1 - \omega_2)(k_1 - k_2)}{k_1 + k_2}$$

$$= 2\hbar\gamma \frac{\cancel{\omega_1 k_1} + \omega_1 k_2 + \omega_2 k_1 + \cancel{\omega_2 k_2}}{k_1 + k_2}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-\frac{(\omega_1-\omega_2)^2}{c^2(k_1+k_2)^2}}} = \frac{c\,(k_1+k_2)}{\sqrt{c^2(k_1+k_2)^2-(\omega_1-\omega_2)^2}}$$

$$\Rightarrow \varepsilon' = 2\hbar c\,\frac{\omega_1 k_2+\omega_2 k_1}{\sqrt{c^2(k_1+k_2)^2-(\omega_1-\omega_2)^2}} = \left(k_j = \frac{\omega_j}{c}\right) =$$

$$= 2\hbar\,\frac{\omega_1\omega_2+\omega_2\omega_1}{\sqrt{(\omega_1+\omega_2)^2-(\omega_1-\omega_2)^2}} = \frac{4\hbar\,\omega_1\omega_2}{\sqrt{4\,\omega_1\omega_2}} = 2\hbar\sqrt{\omega_1\omega_2}$$

$$\varepsilon' = 2\sqrt{\hbar\omega_1\cdot\hbar\omega_2} = 2\sqrt{\varepsilon_1\varepsilon_2}$$

The threshold for electron-positron production is $2m_ec^2 = 10^6\,eV$.

$\Rightarrow$ we need $2\sqrt{\varepsilon_1\varepsilon_2} > 10^6\,eV$

$$\varepsilon_1\varepsilon_2 > (5\cdot 10^5\,eV)^2$$

$$\varepsilon_1 > \frac{(5\cdot 10^5\,eV)^2}{2.5\cdot 10^{-4}\,eV} = \frac{25\cdot 10^{10}}{2.5\cdot 10^{-4}} = 10^{15}\,eV$$

This qualifies photon with ε_1 as a very high energy gamma ray (which are typically $10^{11}-10^{14}\,eV$).